Reg. No.	2	1	1	6	2	3	1	0	0	1	2	5	5

RAJALAKSHMI ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Continuous Assessment Test- I [CAT - I]

Year	: II
Semester	: III
Branch	: B.E Computer Science and Engineering
Sub. Code	: CS23331
Subject Name	: Design and Analysis of Algorithms
QP Code	: 072103

[Regulations 2023]

(Common to Information Technology, Artificial Intelligence and Machine Learning, Computer Science and Design, Artificial Intelligence and Data Science)

Date: 25.09.2024 **Time: 120 Minutes** **Marks: 75**

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

1.1. Define the term "algorithm" and list two characteristics essential for an algorithm. [A1] CO1

1.2. Consider the following algorithm for sorting an integer array using selection sort: [A1] CO1

```
def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i+1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
```

Find the Space Complexity of the Algorithm.

1.3. Given an algorithm that searches for an element in an unsorted array. Find its worst case complexity. [A1] CO1

1.4. Using limits, determine if the function f(n) = 2n + n log n is O(n log n). Provide the steps of your analysis. [A2] CO1

Part B [2 x 11 = 22 Marks]

1.5. a. i)Analyze the time complexity of the following algorithm using Counter method that finds the largest element in an array. [B1] CO1

```
def find_max(arr):
    max_val = arr[0]
    for value in arr:
        if value > max_val:
            max_val = value
    return max_val
```

ii)Give the General Plan to find the time Complexity of Algorithms using mathematical Analysis.

[OR]

b. Explain Big oh and Big Omega Asymptotic Notations in detail with suitable examples. [B2] CO1

1.6. a. Given a set of cities and the distances between each pair of cities, your task is to implement an exhaustive search algorithm to find the shortest possible tour that visits each city exactly once and returns to the starting city. [B1] CO1

	1	2	3	4	5
1	0	10	15	20	25
2	10	0	35	25	30
3	15	35	0	30	20
4	20	25	30	0	15
5	25	30	20	15	0

[OR]

b. Describe a brute force technique to solve the knapsack problem for a set of items with weights [1, 3, 4, 5] and values [1500, 2000, 3000, 4000], with a knapsack capacity of 7. Include a step-by-step explanation of your solution. [B2] CO1

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

2.1. Given the recurrence relation $T(n)=2T(n/2)+n$ solve it using the Master Theorem. [A1] CO2

2.2. State the greedy method abstraction. [A1] CO2

2.3. What is the time complexity of Kruskal's algorithm for finding a minimum spanning tree. [A1]] CO2

2.4. Frame a recurrence relation for the Tower of Hanoi problem and describe the base case and recursive step. [A2] CO2

Part B [2 x 11 = 22 Marks]

2.5. a. Solve the recurrence relation $T(n)=5T(n/5) + n$ and $T(1)=1$ using both backward substitution and the Master Theorem and explain in detail. [B1] CO2

[OR]

b. Given a recursive algorithm for finding the maximum element in an unsorted array of size n, frame the corresponding recurrence relation. Solve the recurrence relation using the forward substitution method and analyze the time complexity. [B1] CO2

```
def find_max(arr, n):
    if n == 1:
        return arr[0]
    return max(arr[n-1], find_max(arr, n-1))
```

2.6. a. You are managing a cargo ship with a maximum weight capacity of 50 units. You have several types of cargo to load onto the ship, each with a specific weight and value: [B1] CO2

- Cargo 1: Weight = 10 units, Value = 60 units
- Cargo 2: Weight = 20 units, Value = 100 units
- Cargo 3: Weight = 30 units, Value = 120 units

Using the Greedy approach for the Fractional Knapsack Problem, determine how you would load the cargo to maximize the total value. Calculate the maximum value that can be obtained and analyze the time complexity of the solution.

[OR]

b. You are scheduling sessions for a conference, and you want to maximize the number of sessions held in a single room without overlap. The sessions have the following start and finish times: [B1] CO2

- Session 1: Start = 2, Finish = 4
- Session 2: Start = 1, Finish = 3
- Session 3: Start = 3, Finish = 5
- Session 4: Start = 0, Finish = 6
- Session 5: Start = 5, Finish = 7

- Session 6: Start = 8, Finish = 9

Apply the Greedy method to select the maximum number of non-overlapping sessions. List the selected sessions and justify your choices. What is the time complexity of the algorithm used?

Answer ALL Questions

Part A [2 x 2 = 4 Marks]

3.1.	Write the recurrence relation for Binary Search Algorithm.	[A1]	CO3
3.2.	State the algorithm to find the minimum and maximum elements in an array using the Divide and Conquer technique.	[A2]	CO3

Part B [1 x 11 = 11 Marks]

3.3.	a.	How does the Divide and Conquer method reduce the time complexity of the Maximum Subarray Problem? Solve the problem for the array [−1,−2,3,4,−5,6,−7] and explain the process.	[B2]	CO3
		[OR]		
	b.	Describe the Divide and Conquer approach. Implement the Binary Search algorithm using this method and explain the step-by-step process with an example.	[B2]	CO3

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Continuous Assessment-I

Computer Science and Engineering

Degree: B. E / B. Tech Semester: III Year: II

Course code & Title: **CS23331 - Design and Analysis of Algorithms**

Date:16.09.2025 QP CODE: 072106

[Common to: IT, AI&ML, CSD, AI&DS]

[Regulations: 2019/2023]

Time: 02.00 hour Max. Marks: 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	Define an algorithm and state two of its key properties.	[CO1]	[k1]
2	List any two basic efficiency classes with one example each.	[CO1]	[k2]
3	What is the order of growth for the function $3n^2 + 4n + 5$?	[CO1]	[k3]
4	State the basic idea of the brute force approach for solving the Knapsack problem.	[CO1]	[k2]
5	Write the general form of a recurrence relation for divide and conquer algorithms.	[CO2]	[k1]
6	Which case in Master Theorem applies when $f(n) = \Theta(n^{\log_b a})$?	[CO2]	[k2]
7	Name any two greedy choice properties used in algorithm design.	[CO2]	[k1]
8	Recall one real-world example where the Huffman coding technique is applied.	[CO2]	[k2]
9	Name the main advantage of the divide and conquer technique.	[CO3]	[k1]
10	What is the time complexity of the binary search algorithm in the worst case?	[CO3]	[k2]

Part B: 5x11=55 Marks

11	a.	i)Describe the counter method for time complexity analysis with a suitable example program. ii)Using limits, compare the orders of growth between $f(n) = 4n^2 + 3n$ and $g(n) = n^3$. Show all calculation steps.	[CO1]	[k3]
		[OR]		
	b.	Define Big-O and Big-Ω notations. Explain their significance in algorithm analysis with suitable examples.	[CO1]	[k3]

12 a. Solve the Travelling Salesperson Problem for the given 4-city distance matrix using the exhaustive search method. [CO1] [k2]

	A	B	C	D
A	0	10	15	20
B	10	0	35	25
C	15	35	0	30
D	20	25	30	0

[OR]

b. Apply the brute force approach to solve the 0/1 Knapsack problem for the given items. Show all combinations and compute total profit. [CO1] [k2]

Item	Weight	Profit
1	2	12
2	1	10
3	3	20
4	2	15

Knapsack capacity = 5

13 a. Using the substitution method, solve [CO2] [k3]

$T(n)=T(n/2)+n^2$

with base conditions T (1) =1 and assuming $n=2^m$. Show the inductive hypothesis and all algebraic steps clearly to obtain a tight bound.

[OR]

b. Explain the different methods for solving recurrence relations in algorithm analysis, including substitution method, Discuss the advantages and limitations of each method with suitable examples. [CO2] [k2]

14 a. A telecom company is planning to lay fibre-optic cables to connect five cities: A, B, C, D, and E. The table below shows the cost (in lakhs) of laying cables directly between each pair of cities. The company wants to connect all cities with the minimum total cost, ensuring that there is a path between every pair of cities without creating any loops. [CO2] [k2]

Using a greedy algorithm, determine the network layout that

achieves this goal. Show all intermediate steps of the algorithm and draw the labelled diagram of the final network.

Connection	Cost (lakhs)
A–B	4
A–C	2
B–C	1
B–D	5
C–D	8
C–E	10
D–E	2

[OR]

b. A thief can carry a maximum weight of **50 kg** in a bag. The following table shows the weights and profits of available items. Using the **Fractional Knapsack** greedy algorithm, determine the maximum profit the thief can earn. Show all intermediate steps, including profit/weight ratio and the fraction of each item taken. [CO2] [k3]

Item	Weight (kg)	Profit (₹)
I1	10	60
I2	20	100
I3	30	120
I4	40	140

15 a. Apply the divide and conquer approach to find the **minimum and maximum** of the given list of numbers: {45, 12, 78, 23, 56, 89, 34, 67}. Show the steps and draw the recursion tree. [CO3] [k3]

[OR]

b. Write and Trace the binary search algorithm for finding the key 56 in the sorted array {12, 23, 34, 45, 56, 67, 78, 89}. Show each mid-comparison step. [CO3] [k3]

0 1 2 3 4 5 6 7